

Research paper

Electricity demand forecasting and power supply planning under carbon neutral targets

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ABSTRACT

The study of electricity demand and power supply planning in accordance with the carbon neutral target is of paramount importance for the attainment of the dual-carbon objective. A system dynamics model that incorporates carbon emissions in order to predict power demand in accordance with the dual-carbon target is proposed. The model incorporates carbon emissions, the output value of three industries, population size, urbanization rate and other factors, and predicts that China's power demand in 2021–2030 will be 1.176×10^{13} kWh. The discrepancy between the model and the prediction of the China Electric Power Research Institute (CEPRI) is 0.3%. Subsequently, a power supply planning model that considers the dual objectives of comprehensive cost and carbon emission is constructed, with the parameters set based on the current status of China's power supply structure, in conjunction with relevant policies and future development trends. The model is ultimately solved using the non-dominated sorting genetic algorithm II (NSGA-II) coupled entropy weight Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) method, and the optimal compromise is determined on the Pareto frontier. The total cost of adopting the power structure proposed in this paper is estimated to be approximately RMB 6.86 trillion, with a corresponding CO₂ emission of 4.529 billion tons. It is projected that the proportion of non-fossil energy sources in the installed capacity will reach 61.13% by 2030, comprising 1.214 billion kW of wind and photovoltaic, 412 million kW of hydroelectricity, and 220 million kW of nuclear power. The findings of this study can inform the future development of China's power industry, particularly with regard to the evolution of its energy structure and the refinement of its low-carbon transition pathway.

1. Introduction

At present, the carbon emissions of China's coal-based power industry account for more than 40% of the total social carbon emissions (Wan et al., 2021), and in order to achieve the dual-carbon target as scheduled, the carbon emission reduction of the power industry should be the first to bear the brunt. Establishing a new power system with a high proportion of new energy is both a challenge for the power industry and an opportunity for the strategic transformation of energy structure. Therefore, it is crucial to formulate practical power structure planning, and accurate prediction of power demand under the dual-carbon target is an important prerequisite for scientific planning of power structure.

For the research of electricity demand forecasting, scholars have proposed methods such as integrating moving average autoregressive model (Ahmad et al., 2019), neural network (Bot et al., 2020), grey

forecasting (Fan et al., 2021), and non-parametric auto regression (Lindberg et al., 2019). Literature (Zhang et al., 2023) proposed a method for ultra-short-term load forecasting based on Variational Mode Decomposition and Multi-model Fusion, which combines the loads in the integrated energy system with the meteorological information, and uses three different machine learning models to make forecasts, and then fuses the forecasts to get the final results through support vector machines. Aiming at the characteristics of large amount of historical data and few data features in power load forecasting, a two-layer XGBoost algorithm is proposed in the literature (Gao et al., 2022), which can predict the ultra-short-term power loads with high accuracy and good generalization. Based on extreme gradient boosting and time convolution network, literature (Suneetha Emmela et al., 2024) proposed a short-term power load forecasting method, which can effectively deal with the impact of short-term load fluctuation, stochastic and

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other characteristics. Based on extreme gradient boosting and time convolutional network, literature (BASHIRI et al., 2023) incorporated temperature into the short- and medium-term power load forecasting model, comparing the neural sequence and feed-forward neural network models before and after the introduction of temperature. Literature (Brusafferri et al., 2022) developed a short-term power load forecasting method based on Bayesian optimization of convolutional neural network-bidirectional gated recurrent network, which can fully exploit the temporal characteristics of load data. Literature (WU et al., 2018) used improved grey correlation analysis to select similar days in short-term electricity load forecasting and optimized BP neural network by bat algorithm for short-term load forecasting. Literature (Wang Zhengxin et al., 2018) proposed a seasonal grey model based on the cumulative operator to forecast the electricity consumption of the primary industries in Chinese regions. Starting from the influencing factors of electricity demand in three time scales: daily, monthly and yearly, literature (Abbasimehr et al., 2020) used long and short term memory networks to forecast the medium and long term loads in the region.

Most of the studies on medium- and long-term load forecasting are based on intelligent algorithms, machine learning and other methods, while medium- and long-term load forecasting requires not only the extraction of the pattern of change of historical data, but also the consideration of regional development trends. The prediction accuracy of methods such as machine learning is high, but the physical model of power demand is huge, and the use of intelligent algorithms and machine learning methods is unable to analyze the system qualitatively and quantitatively, and cannot explore the comprehensive impact of its internal mutual constraints and interaction mechanisms on the external system.

Meanwhile, power planning can be divided into grid planning and power supply planning, but scholars have less research on power structure planning. Since the 1960s, researchers have developed many power planning optimization and decision-making software with various characteristics (Deng and Lv, 2020). On this basis, domestic scholars have also begun the study of power supply planning models. Literature (Zhou et al., 2024) establishes a two-tier power planning model with coordinated wind-fire ratios taking into account capacity tariffs, which provides a reference for the synergistic development of future power planning and market mechanisms. Literature (Ma et al., 2024) established a new energy and energy structure optimization model with the objective function of minimizing power generation cost and pollutant emission. Literature (Manesh et al., 2024) establishes a power supply planning model taking into account the coal power and water footprint. Literature (Scuzziato et al., 2021) developed a solution method with alternating long and short time dual-scale running iterations, and verified the reliability of the model using the provincial power system as an example. Literature (Wang et al., 2023) developed a planning model for power structure with multi-stage stochastic planning and multi-power structure planning, but the model did not involve carbon emission constraints, and could not guarantee that the power structure could meet China's dual-carbon target of 30–60.

Since it is impossible to quantitatively characterize the relationship between carbon emission constraints and power supply structure under the dual-carbon target, measuring the power demand under the dual-carbon target becomes a difficult problem, and most of the existing power supply planning research focuses on reducing the investment and construction costs and operation costs, without considering the carbon constraints, and does not involve the electric power loads under the constraints of the dual-carbon target, so that it is not possible to give the reasonable power supply structure under the carbon-neutral target in China.

To solve this situation, a dual-carbon target-oriented optimization method of power supply structure is proposed, which adopts a simple and feasible physical model instead of the original software model, avoiding the disadvantages of using software to measure the huge amount of data required, complex data statistics and difficult scenario

analysis; To avoid the shortcomings of machine learning and other methods for medium and long-term power demand forecasting, a system dynamics (SD) model with carbon emission constraints was established, and a scenario in which China's carbon emissions peak by 2030 was introduced as a constraint, and the power demand under the constraints of the dual-carbon target was obtained; With this power demand as a constraint, it was introduced into the power supply planning model with the objective of minimizing the total cost of the system and the least amount of carbon emission, and the Pareto II model was solved using NSGA-II to obtain the power demand under the constraints. NSGA-II is used to solve the Pareto frontier, and the entropy weight TOPSIS method is used to rank the solutions and select the power planning scheme with the best overall performance.

The novelty of this paper is as follows: First, a system dynamics model that considers carbon emissions is proposed to forecast electricity demand under the dual carbon target, taking into account factors such as carbon emissions, output value of three industries, population size and urbanization rate. Then, a power planning model is developed that takes into account both integrated cost and carbon emission targets. No similar studies were identified in the available literature.

2. System dynamics modelling of electricity demand

2.1. Analysis of factors affecting electricity demand

In the process of applying system dynamics for electricity demand forecasting, individual influencing factors that have direct or indirect correlation with electricity demand are identified. In this study, the following major factors are considered to ensure the accuracy and reliability of the forecasts:

1) Level of economic development

Gross domestic product (GDP) growth is usually accompanied by an expansion of industrial and commercial activities, which in turn drives up the demand for electricity. As the economy continues to grow, the demand for lighting, air-conditioning and other household appliances increases, leading to a rise in household electricity consumption.

2) Urbanization process

Increased levels of urbanization mean higher population densities and a concentration of industrial and commercial activities, which require more power support to meet the demands of daily life and economic activities.

3) Population Factors

Population growth directly increases the demand for residential, commercial and industrial, and utility power. The rate of population growth and its size are key factors that influence the amount of electricity demanded.

4) Electricity consumption structure

In the power system, the total power consumption structure can be subdivided into industrial power consumption and social power consumption. Specifically, the former includes the power demand of three major sectors: primary, secondary, and tertiary. In addition, the latter is further divided into urban and rural electricity consumption.

5) Industrial structure

The demand for electricity varies significantly from industry to industry, especially in the secondary industry, where electricity consumption and intensity are usually higher than that of the primary and

tertiary industries due to the reliance on high-power equipment for industrial production and machining processes.

6) Carbon emissions

The impact of carbon emissions on electricity demand is mainly reflected in the constraints of the dual-carbon target, which is mainly reflected in the promotion of energy structure optimization, energy efficiency, electrification demand growth, and environmental policy adjustments.

7) Energy structure

The energy structure will directly affect electricity demand, such as increasing the proportion of renewable energy, reducing the use of coal and other fossil fuels, improving energy efficiency and meeting the demand for electrification, etc. will have a profound impact on the scale of electricity demand and mode of supply.

2.2. Causal loop diagram for electricity demand system

Causal loop diagram is a central tool in system dynamics for revealing the interactions and feedback mechanisms between the constituent elements within a system. In the comprehensive system of electricity demand, the system behavior is influenced by numerous internal and external factors. By constructing causal loop diagrams, the challenges faced by the power industry can be systematically analyzed.

Fig. 1 illustrates the four key feedback loops in the electricity carbon system, analyzing their role in the system dynamics and their impact on the overall carbon trend.

- (1) Negative feedback between economic growth and carbon emissions: $GDP \rightarrow + \text{consumption level} \rightarrow + \text{urbanization rate} \rightarrow + \text{demand for electricity} \rightarrow + \text{generation of electricity} \rightarrow + \text{consumption of fossil energy} \rightarrow + \text{carbon emissions} \rightarrow + \text{cost of abatement} \rightarrow - GDP$.

Living standard Rising is contributing to urbanization, leading to a significant increase in electricity demand in towns and cities. As the urbanization rate increases, the overall electricity consumption grows, leading to a rise in the electricity demand, which in turn increases the consumption of fossil fuels and carbon emissions from the power sector.

In this process, the increase in carbon emissions triggers a rise in the cost of emission reduction, which in turn has a negative impact on economic development, forming a negative feedback.

- (2) Positive feedback between economic and technological development: $GDP \rightarrow + \text{investment in technology} \rightarrow + \text{scientific and technological progress} \rightarrow + GDP$.

Rapid economic growth incentivizes more investment in technological research and development, and technological upgrading in turn contributes to economic growth, creating a positive enhancement. Technological advances not only increase productivity, but also help to develop cleaner and more efficient ways of using energy, thus contributing to economic growth while reducing carbon emissions.

- (3) Negative feedback between carbon emissions and environmental protection: $\text{carbon emissions} \rightarrow - \text{environmental quality} \rightarrow + \text{government intervention} \rightarrow + \text{environmental protection investment} \rightarrow + \text{carbon sinks} \rightarrow - \text{carbon emissions}$.

The increase in carbon emissions from the power sector has worsened environmental quality, prompting the government to increase the strength of its environmental policy and investment in environmental protection, and to reduce carbon emissions through the implementation of carbon sink technologies such as carbon capture, utilization and sequestration (CCUS), thus creating a negative feedback loop that helps to slow down the growth of carbon emissions.

- (4) Negative feedback between new energy development and carbon emission reduction: $\text{carbon emissions} \rightarrow - \text{environmental quality} \rightarrow + \text{government intervention} \rightarrow + \text{technology investment} \rightarrow + \text{new energy utilization} \rightarrow - \text{fossil energy consumption} \rightarrow - \text{carbon emissions}$.

The increase in carbon emissions from electricity has led to a decline in environmental quality, increasing pressure on governments to reduce emissions and promoting the introduction of policies to encourage technological innovation and new energy development. By promoting the development and application of grid-connected renewable energy technologies, these policies reduce reliance on fossil fuels and lower carbon emissions, creating a negative feedback loop that helps to achieve a low-carbon transition in the power sector.

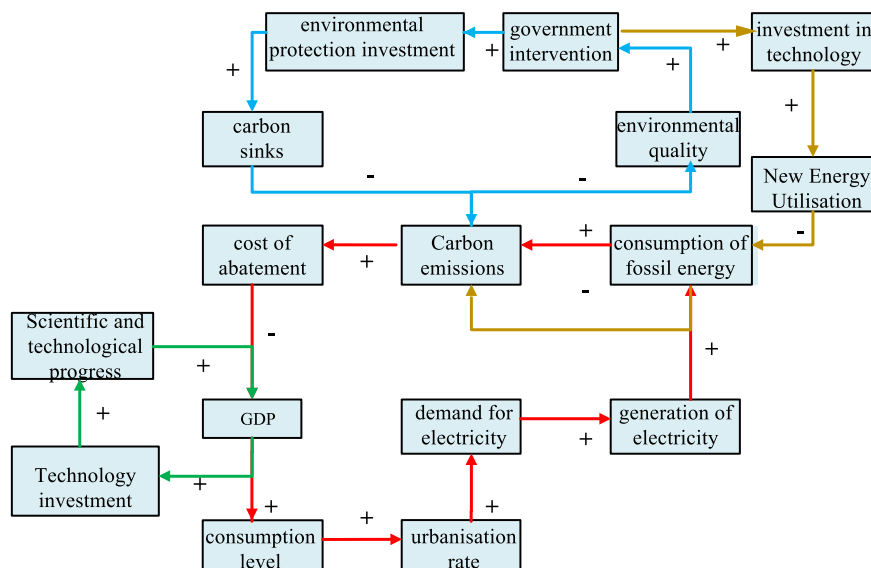


Fig. 1. Electricity Demand System Cause and Effect Diagram.

2.3. System stock flow diagram for electricity demand

The construction of a system dynamics model involves several elements, including state variables, rate variables, auxiliary variables and constants, and aims to describe the dynamics within the system and its interaction with the outside world. In this paper, the electricity demand forecasting model is composed of the following key variables: gross domestic product, total population, level of urbanization, electricity intensity of primary to tertiary industries, industrial structure, total carbon emissions and total electricity demand. The model is further refined into three sub-systems: the primary to tertiary electricity use system, the carbon emissions sub-system and the urban and rural electricity use system, as shown in Fig. 2. The equations for the main variables in the model are from Reference (Wu et al., 2024).

(1) Three industrial electronic systems

Electricity consumption in the tertiary sector is jointly affected by the industrial output value and the intensity of electricity consumption, where the industrial output value is closely related to the GDP and its relative share in the national economy. In addition, the impact of carbon emission constraints on the share of the secondary sector is particularly considered, and a positive correlation is found between the two. The urbanization process also has an indirect impact on the intensity of electricity consumption in the tertiary sector, especially the close correlation between the urbanization rate and the electricity consumption of the tertiary sector. The data are obtained from the National Bureau of Statistics.

(2) Urban and rural electricity consumption system

The urban-rural electricity consumption structure covers the electricity consumption of urban and rural residents, and is calculated based on the respective population size and per capita electricity intensity. Using the 2003 population data as a benchmark, the natural population growth rate and total population are applied to make projections and further calculate the population size in future years, with the urbanization rate being the key variable in determining the size of the urban

and rural population. This part of the data is quoted from the National Bureau of Statistics and the China Energy Administration, which provides data support for the analysis of urban and rural electricity consumption.

(3) Carbon emission subsystems

According to the analysis of literature (Sun Meng et al., 2023), the carbon emissions level is influenced by several key factors such as energy structure, population size, energy intensity, GDP per capita and urbanization rate. The impact of energy structure is quantified by assessing the share of coal and other fossil energy sources in total energy consumption, while energy intensity is measured by the relationship between total energy consumption and GDP. These factors are crucial for developing effective carbon emission reduction strategies and promoting environmental sustainability. Relevant data were obtained from the literature (Sun Meng et al., 2023) and the National Bureau of Statistics.

2.4. Validity of the proposed model

The constructed SD model is tested by the historical review test method, and the electricity demand is selected as the model test variable. Statistical data from 2003 to 2014 are used to train the model, and the total electricity consumption from 2015 to 2019 is obtained by solving the model as a control variable (the data are all taken from the National Bureau of Statistics (NBS), because the total energy consumption by province is only counted until 2019 in the NBS, and the other data are counted until 2022, thus 2015–2019 is selected as the test set in this paper), and the training and testing ratio of the model is 12:5. The training-testing ratio of the model is 12:5. Carbon emissions are calculated from the 2006 IPCC Guidelines for National Greenhouse Gas Inventories and the default CO₂ emission coefficients of 8 types of fossil fuels provided by the National Bureau of Statistics (NBS), and the provincial consumption of fossil fuels for energy are used to calculate the annual CO₂ emissions of each province respectively, and the specific calculation methods are described in the literature (Sun Meng et al., 2023).

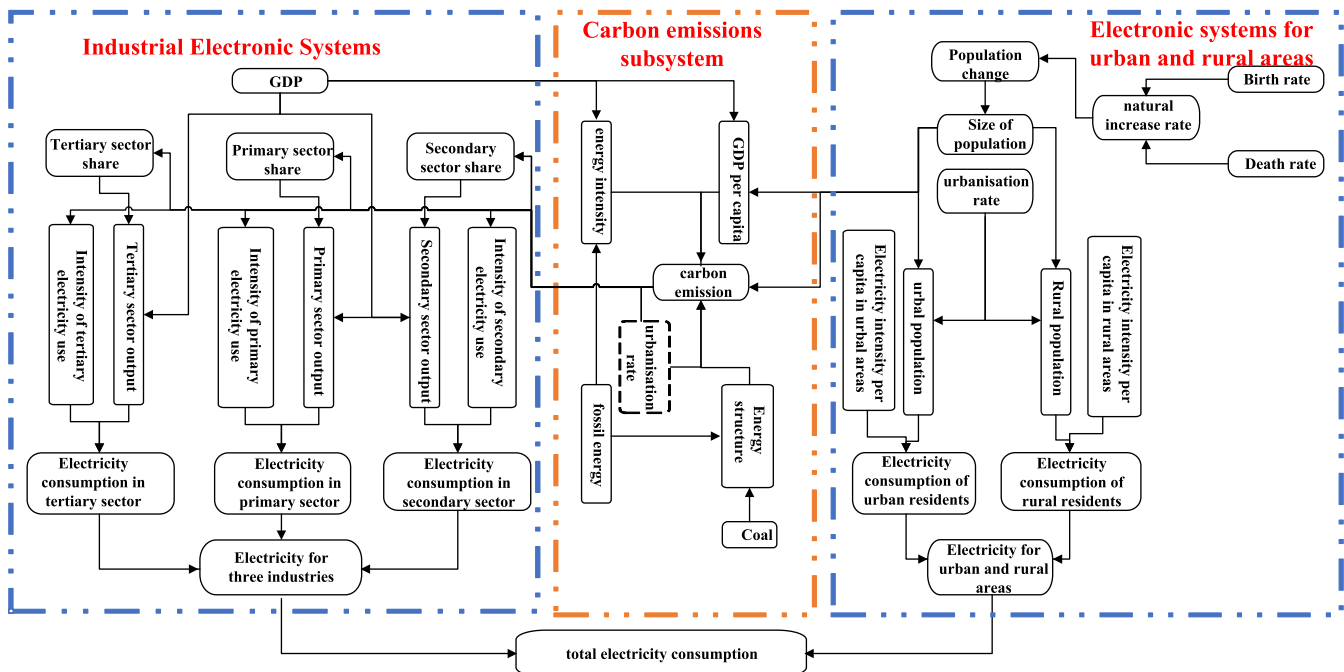


Fig. 2. Electricity demand system stock flow diagram.

In this paper, the accuracy of the model is measured by the mean absolute percentage error e_{MAPE} shown in Eq. (1).

$$e_{MAPE} = \frac{1}{N} \sum_{l=1}^N \left| \frac{y_l^* - y_l}{y_l} \right| \times 100\% \quad (1)$$

Where y_l and y_l^* are the real and predicted values of electricity demand in year l , respectively; N is the number of years. The detection results of the SD model of electricity demand from 2003 to 2019 are shown in Fig. 3.

As can be seen from Fig. 3, the e_{MAPE} of the model training set and test set are 1.73 % and 1.71 %, respectively, which are significantly less than 5 %, indicating that the model prediction accuracy is qualified. Among them, the maximum error of the test set is $2.59 \% < 3.5 \%$, and the prediction accuracy is within the excellent range (Chen and Zhou, 2014) (prediction error: $< 3.5 \%$ is good, $< 5 \%$ is qualified, $< 10 \%$ is barely qualified, and $\geq 10 \%$ is unqualified), which indicates that the SD model established in this paper can effectively reflect the actual power demand system and has high reliability.

2.5. Electricity demand forecast

The parameter settings of the electricity demand forecasting scenarios adopt the two peak carbon scenarios of high-speed and medium-speed from the literature (Sun Meng et al., 2023), which are denoted as Scenario I and Scenario II, and the two scenarios can achieve different degrees of peak carbon in 2028 and 2029, respectively, and the specific data are shown in (Sun Meng et al., 2023). In this paper, Scenario III, which does not consider the carbon constraints, is introduced as a control group.

The forecast values of electricity demand for the three scenarios are given in Fig. 4. In Fig. 4, the e_{MAPE} for scenario III in 2003–2019 is 2.27 %, which is higher than the 1.72 % for scenarios I and II, indicating that the introduction of carbon emission variables in the SD improves the prediction accuracy of the model. Scenarios I and II are two carbon peaking scenarios and their electricity demand in 2030 is 1.176×10^{13} kWh and 1.202×10^{13} kWh, respectively. If carbon emission constraints are not taken into account, it will make the electricity demand grow rapidly following the development of the economy, and thus the electricity demand in 2030 in Scenario III reaches 1.340×10^{13} kWh.

Table 1 gives the projected values of electricity demand in 2030 from the existing literature. Among them, Y. Shu, academican of the Chinese Academy of Engineering, gives a roughly 1.18×10^{13} kWh for China's electricity demand in 2030 under the dual-carbon target (Shu et al., 2021). Comparison reveals that the other literature predicts electricity demand without taking into account the carbon emission constraints

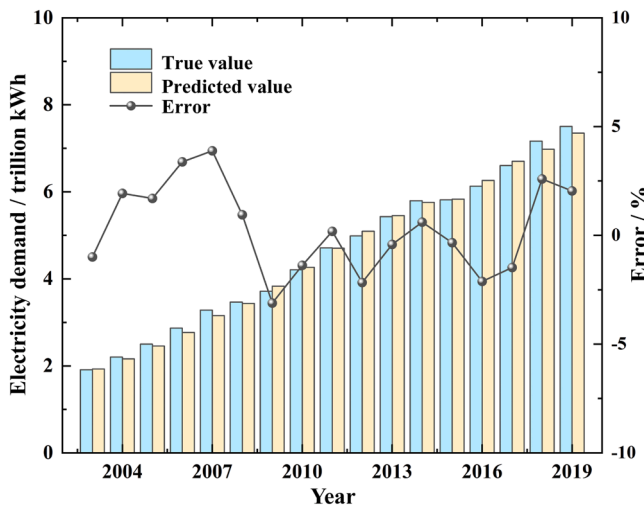


Fig. 3. Electricity demand SD model historical review test results.

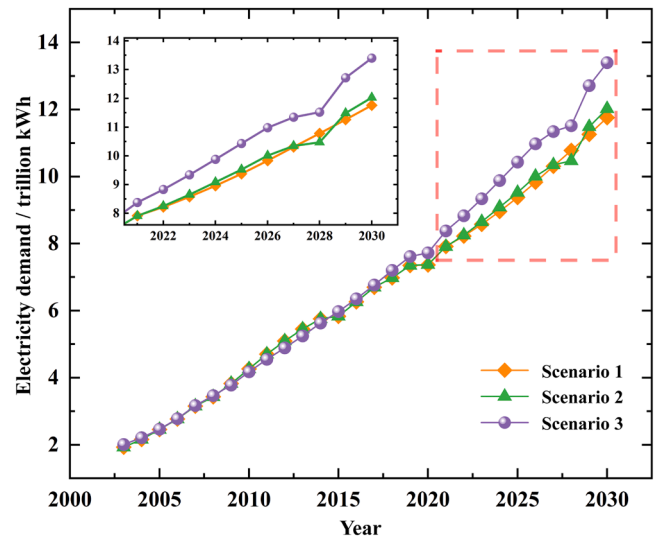


Fig. 4. Electricity Demand Three Scenario Forecast Results.

Table 1

Comparison of predicted results in the literature.

literature	Prediction Model	Projected results for 2030
(Shu et al., 2021)	-	1.180×10^{13} kWh
(Lei et al., 2020)	Kaya identity	0.963×10^{13} kWh
(Luo and Zhang, 2024)	Ridge Regression	1.013×10^{13} kWh
(Yuan et al., 2023)	"S" trajectory	1.130×10^{13} kWh
(Liu et al., 2022a)	-	1.044×10^{13} kWh
(Yao et al., 2021)	-	1.060×10^{13} kWh
(Li et al., 2022)	SD	1.171×10^{13} kWh
This work	SD	1.176×10^{13} kWh

and with fewer factors considered in the explanatory variables, which leads to a large difference between their predictions and the results of the literature (Shu et al., 2021). In contrast, the factors considered in Scenario 1 of this paper are more comprehensive and incorporate carbon emission constraints, and the predicted results differ from literature (Shu et al., 2021) by 0.3 %. Literature (Li et al., 2022) also adopts a system dynamics model and introduces carbon emission factors, and its predicted results differ from literature (Shu et al., 2021) by 0.7 %. Literature (Li et al., 2022) introduces all carbon emission factors and related policies into the model, which makes the model complex and the setting of model parameters is relatively difficult, this paper simplifies the model on its basis, and only introduces the key factors affecting the carbon emission constraints into the model, and the parameters related to the carbon emission are parameterized on the basis of the peak already reached in the literature (Sun Meng et al., 2023), which makes the model more streamlined. The above comparison shows both the validity and reliability of this paper's model, and also supports the necessity of considering carbon emissions in electricity demand forecasting.

3. Power planning modelling

3.1. Objective function

The objective of the power supply planning model in this work is to minimize the total cost of power generation and minimize CO₂ emissions. The power source structure is eight kinds of coal power, gas power, hydropower, photovoltaic power, wind power, nuclear power, biomass power, pumped storage and so on.

1) Total cost of power generation

The total cost of power generation f includes four parts: investment and construction cost of new units, unit operation cost, carbon emission cost, and revenue from electricity sales, as shown in Eq. (2).

$$\min f = \sum_{i=1}^U \sum_{t=1}^T (u_{i,t} \Delta Q_{i,t} P_{i,t} + H_{i,t} Q_{i,t} \delta_{i,t} - H_{i,t} Q_{i,t} S_{i,t}) + f_{CO_2} \quad (2)$$

Where U is the set of planned power supply; T is the time of power supply planning; $u_{i,t}$ indicates whether there is an addition of the i th power supply in year t ; $\Delta Q_{i,t}$, $P_{i,t}$, $H_{i,t}$, $Q_{i,t}$, $\delta_{i,t}$, and $S_{i,t}$ are the new installed capacity of the i th power supply in year t , the cost of investment and construction of the capacity per unit, the number of hours of annual utilization, the total installed capacity at the end of the year, the operation and maintenance costs, and the selling price of electricity, respectively; f_{CO_2} is the cost of carbon emissions, as shown in Eq. (3).

$$f_{CO_2} = \sum_{i=1}^2 \sum_{t=1}^T P_{i,t}^{CO_2} \cdot k_{i,t}^{CO_2} \cdot H_{i,t} \cdot Q_{i,t} \quad (3)$$

Where $P_{i,t}^{CO_2}$ denotes the carbon emission cost of the i th power source in year t ; $k_{i,t}^{CO_2}$ denotes the CO_2 emission intensity of the i th power source in year t .

2) CO_2 emission

$$\min E_{CO_2} = \sum_{i=1}^2 \sum_{t=1}^T k_{i,t}^{CO_2} \cdot H_{i,t} \cdot Q_{i,t} \quad (4)$$

Where E_{CO_2} is CO_2 emission.

3.2. Constraints

The constraints mainly contain electric power balance constraints, carbon and pollutant emission constraints, installed capacity constraints, power plant profitability constraints, energy saving and emission reduction constraints, etc., which are shown in Eqs. (5) to (12), respectively.

1) Power balance constraints

$$\sum_{i=1}^N \sum_{t=1}^T Q_{i,t} \geq L_t^{\max} (1 + \gamma_t) \quad (5)$$

Where $L_t^{\max}$ denotes the maximum load in year t ; γ_t is the spare capacity factor of the system equipment during the planning period in year t .

2) Balance of power constraints

$$\sum_{i=1}^N \sum_{t=1}^T H_{i,t} \cdot Q_{i,t} \cdot (1 - \theta_t^{\text{loss}}) \geq D_t^{\max} \quad (6)$$

Where θ_t^{loss} denotes the combined plant electricity consumption rate and network loss rate in year t ; $D_t^{\max}$ denotes the electricity demand in year t .

3) Carbon emission constraints

$$\sum_{i=1}^2 \sum_{t=1}^T k_{i,t}^{CO_2} \cdot H_{i,t} \cdot Q_{i,t} \leq E_{CO_2,t}^{\max} \quad (7)$$

Where $E_{CO_2,t}^{\max}$ denotes the permitted CO_2 emission limit for year t .

4) Pollutant emission constraints

$$\begin{aligned} \sum_{i=1}^2 \sum_{t=1}^T k_{i,t}^{SO_2} \cdot H_{i,t} \cdot Q_{i,t} &\leq E_{SO_2,t}^{\max} \\ \sum_{i=1}^2 \sum_{t=1}^T k_{i,t}^{NO_x} \cdot H_{i,t} \cdot Q_{i,t} &\leq E_{NO_x,t}^{\max} \\ \sum_{i=1}^2 \sum_{t=1}^T k_{i,t}^{YC} \cdot H_{i,t} \cdot Q_{i,t} &\leq E_{YC,t}^{\max} \end{aligned} \quad (8)$$

Where $k_{i,t}^{SO_2}$, $k_{i,t}^{NO_x}$, $k_{i,t}^{YC}$ denote the emission intensity of SO_2 , NO_x and soot, respectively; $E_{SO_2,t}^{\max}$, $E_{NO_x,t}^{\max}$, $E_{YC,t}^{\max}$ denote the permissible emission limits for SO_2 , NO_x and soot in year t , respectively.

5) Installed capacity constraints

$$Q_{i,t}^{\min} \leq Q_{i,t} \leq Q_{i,t}^{\max} \quad (9)$$

Where $Q_{i,t}^{\max}$ and $Q_{i,t}^{\min}$ denote the upper and lower bounds on the installed capacity of the i th power source in year t , respectively.

6) Power plant profitability constraints (new power installations become profitable over their lifetime planning period)

$$\sum_{i=1}^U \sum_{t=1}^T \Delta Q_{i,t} \cdot P_{i,t} + (H_{i,t} \Delta Q_{i,t} \delta_{i,t} - H_{i,t} \Delta Q_{i,t} S_{i,t}) \cdot B_i \geq 0 \quad (10)$$

Where B_i denotes the life planning cycle of the i th power supply.

7) Clean energy installed capacity constraints

$$\sum_{i=2}^N \sum_{t=1}^T Q_{i,t} \geq Q_t^{\min, c} \quad (11)$$

where $Q_t^{\min, c}$ denotes the lower limit of installed clean energy capacity in year t .

8) Energy efficiency and emission reduction constraints (the key point of energy efficiency and emission reduction performance is the reduction of emissions per unit of electricity generation, i.e. the reduction of its emission intensity σ_t^{Em} in year t compared to year $t-1$)

$$\frac{\sum_{i=1}^2 \sum_{t=2}^T k_{i,t}^{\text{Em}} \cdot H_{i,t} \cdot Q_{i,t}}{\sum_{i=1}^2 \sum_{t=2}^T H_{i,t} \cdot Q_{i,t}} \leq 1 - \sigma_t^{\text{Em}} \quad (12)$$

Where $k_{i,t}^{\text{Em}}$ denotes the emission intensity of the i th power source in

year t , including CO₂, SO₂, NO_x, and soot.

9) Coal power decommissioning limit constraints

The annual stranded capacity of coal power units should not be too large, and for this purpose, it is set to take 10 per cent of the total existing capacity as the upper limit of the annual stranded size of the units.

$$\sum_{i=1}^2 \sum_{t=1}^T \Delta Q_{it} \leq \sum_{i=1}^2 \sum_{t=1}^T Q_{it} \times 10\% \quad (13)$$

3.3. Determination of the main parameters of the model

3.3.1. Cost parameters and learning progress assumptions

In the field of machine learning, the learning rate is a key parameter used to control the step size of parameter updates during the training process of a model, which directly affects the magnitude of parameter updates along the gradient direction in each iteration. The performance of model training can be effectively improved by dynamically adjusting the learning rate during training, i.e., implementing a learning rate scheduling strategy. In this paper, this concept is applied analogously to the analysis of power supply investment costs. By introducing the concepts of the number of learning progresses and the learning rate, the trends of the costs of different power supplies over time can be quantified. That is, the time-varying relationship between construction and operating costs is linked to the learning progress number and learning rate as a means of characterizing the cost reductions brought about by various power supply technologies over time and technological advances. In this paper, the number of learning progress and the corresponding learning rate are set in conjunction with the predictions of industry experts on the cost reduction of different power sources (Lei et al., 2020), as shown in Table 2.

In the power industry, the components of power generation operating costs include, but are not limited to, fuel costs, maintenance costs, insurance costs and employee salaries. In particular, for renewable energy generation methods such as wind, solar and hydro, their operating costs are relatively low as they require no fuel. On the contrary, coal-fired and natural gas-based thermal power generation units require a large amount of fuel during operation, and their operating costs are on a significant upward trend with the increase in coal prices and the technological modifications required to meet the ultra-low emission standards. In addition, biomass power generation also faces high operating costs due to its high fuel and storage costs and relatively low generation efficiency. In this paper, the learning rate is used to measure the ability of technology costs to decline over time due to economies of scale and technological advances. In this paper, the learning rate of operating costs for coal power exceeds 1 (Lei et al., 2020), reflecting an increasing trend in operating costs for coal power instead, due to higher coal prices and increased investment in environmental improvements. In contrast, renewable energy technologies, such as wind and solar, typically have learning rates below 1, implying cost reductions over time, due to increased technological maturity and production scale-up. The operating cost and learning rate parameters are shown in Table 2.

Table 2

Power investment construction costs and learning rates.

Power source	Base price /(¥/kW)	Learning Progress			learning rate
		2020	2025	2030	
coal power	3800	2	2	2	90 %
gas power	4000	2	4	6	85 %
hydropower	10000	1	1	1	90 %
wind power	7200	2	4	6	90 %
photovoltaic	8000	2	4	6	85 %
nuclear power	13000	3	6	11	90 %
Biomass power	9000	2	4	6	85 %
pumped storage	4500	1	1	1	90 %

The operating cost and learning rate parameters for each power supply are shown in Table 3.

Based on the predictions in the *China Carbon Price Survey Report 2020* and the actual carbon market quotes in 2021, the carbon price in 2021 is set at 53 ¥/ton, and is expected to reach 150 ¥/ton by 2030 and rise to 300 ¥/ton by 2060, which is based on the current average carbon price in the carbon markets of major countries and regions around the world. Over the planning period, it is assumed that the carbon price will follow a linear growth pattern year on year.

3.3.2. Pollutant emission parameters

The *Action Plan for Energy Saving, Emission Reduction, Upgrading and Retrofitting of Coal-fired Power Plants* states that by 2020, all coal-fired power plants with the ability to be retrofitted across the country will achieve ultra-low emissions, with their soot, SO₂, and NO_x emission concentrations no higher than 10, 35, and 50 mg/m³, respectively. The IEA, on the basis of the current state of technological development of coal-fired power generation, proposes the pollutant emission targets for coal-fired power plants by 2030: soot < 1, SO₂ < 10, NO_x < 10 mg/m³ (Liu et al., 2022b). The ‘14th Five-Year Plan’ for modern energy system specifies that the average emission concentrations of soot, SO₂, and NO_x in the national thermal power industry will be 5, 30, and 45 mg/m³, respectively, in 2025, and that the values will be reduced to 3, 25, and 35 mg/m³, respectively, in 2030. intensity refer to the Study on Coal Consumption Control Programme and Policies in Electric Power Sector, as the intensity of soot emission is relatively small compared to other pollutant emissions (Wei et al., 2024), soot emission is assumed to account for 100 % of the capacity of ultra-low-emission coal power generation units in 2025 and 2030, and the specific values of CO₂ as well as other pollutant emission intensities are shown in Table 4.

3.3.3. Feed-in tariffs and life cycle

The *Notice of the National Energy Administration on Doing a Good Job in Signing and Complying with Medium- and Long-Term Contracts for Electricity in 2023* states that the national coal feed-in tariff should implement the requirements of the ‘benchmark price + up and down fluctuation’ market-based price mechanism. For new energy power generation, the NDRC introduced a new energy feed-in tariff policy in 2021, explicitly requiring that the feed-in tariffs for newly filed centralized PV power stations and distributed PV projects and onshore wind power projects be implemented in accordance with the benchmark price of local coal prices, and that the ‘feed-in tariff’ policy in 2022 provided strong support for the rapid development of new energy. In 2022, the policy of ‘grid parity’ was clarified to provide strong support for the rapid development of new energy sources, and the parameters were set as shown in Table 5 (Lei et al., 2020).

3.3.4. Incremental parameter of installed capacity

China’s ‘14th Five-Year Plan’ development plan and the ‘*China Energy Outlook 2030*’ proposed to optimize the layout and structure of energy production, increase the development of green energy, enhance the construction of energy reserve facilities, and strive to enhance the energy emergency peak reserve capacity to strengthen the construction of

Table 3

Power supply running costs and learning rates.

Power source	Base price /(¥/kWh)	Learning Progress			learning rate
		2020	2025	2030	
coal power	0.25	2	4	6	115 %
gas power	0.35	1	1	1	90 %
hydropower	0.02	1	1	1	90 %
wind power	0.08	1	1	1	90 %
photovoltaic	0.10	5	15	23	70 %
nuclear power	0.10	1	1	1	90 %
Biomass power	0.40	1	1	1	90 %
pumped storage	0.02	1	1	1	90 %

Table 4
Pollutant and greenhouse gas emission intensity (g/kWh).

		2020	2025	2030
Coal	CO ₂	865	855	850
	SO ₂	0.6	0.4	0.4
	NOx	0.8	0.6	0.6
	soot	0.036	0.015	0.009
Gas	CO ₂	515	515	515
	NOx	0.4	0.4	0.4

Table 5
Power life expectancy and feed-in tariffs.

Power source	Life expectancy /Year	feed-in tariff/ (¥/kWh)		
		2020	2025	2030
coal power	30	0.38	0.38	0.38
gas power	25	0.42	0.42	0.42
hydropower	40	0.27	0.27	0.27
wind power	20	0.50	0.50	0.50
photovoltaic	15	0.82	0.78	0.76
nuclear power	50	0.45	0.47	0.45
Biomass power	15	0.60	0.58	0.55
pumped storage	40	0.25	0.25	0.25

the energy emergency reserve system, and enhance the ability to ensure the security of energy supply. Promote energy integration and optimization, and promote the installed capacity of new energy sources such as wind power and photovoltaic power generation. Vigorously develop natural gas distributed energy, and increase efforts to promote the development of natural gas power generation and wind power, solar power, biomass power generation, energy storage and other new energy sources in-depth integration. On the basis of the relevant policies, this paper combines the relevant data of the Chinese Academy of Engineering, the parameters of the incremental installed capacity size set are shown in Table 6.

4. Case study

4.1. Assumptions

In order to enhance the practical application value of the model, this study sets up the model based on a series of assumptions, and the main assumptions involved include:

(1) Single-node concept.

In view of the differences in focus and optimization difficulty between power planning and grid planning, this study adopts the single-node power planning concept, i.e., all loads and generating units in the power system are regarded as being brought together in a virtual node, which is similar to the power planning procedure of the International Atomic Energy Agency.

It is necessary to point out that power planning needs to deal with

Table 6
Installed size increment parameterization.

Power source	Incremental installed capacity cap/ (10 MW)	
	2021–2025	2026–2030
coal power	3000	1000
gas power	1400	1400
hydropower	1400	700
wind power	6000	7000
photovoltaic	7000	7000
nuclear power	1000	1400
Biomass power	1000	1200
pumped storage	1400	1400

various types of generating units and requires consideration of the transition of system power over long periods of time. This results in a problem involving a very large number of decision variables. The single node model assumes that the entire load of the system with all the generating units is considered to be concentrated at a single node. With this assumption, it is not necessary to consider the geographic distribution of power sources and electrical loads for power planning, so that similar generating units can be combined for optimization, making the model's algorithms much simpler. However, this model cannot answer the question of where to expand these units, and its error may be larger for countries with large geographical areas and uneven energy distribution.

(2) Constant project development cycle.

It is assumed that the entire cycle time from planning, construction to operation of a power project is fixed, and is not affected by factors such as project size, location, environmental approvals, or financial planning, although the actual situation may vary.

At the same time, the assumption of a constant development cycle for the project ensures that all investment costs, residual value of investment, etc., are essentially constant, thus avoiding the complexity of the problem due to the stochastic nature of fuel and equipment prices.

(3) Stable policies and regulations. It is assumed that government policies, laws and regulations remain stable within the timeframe of power supply planning, without considering the impact of possible future policy changes on power system planning.

(4) Diversified power system form. It is predicted that the future power system will strengthen the development of both centralized and distributed systems, with special emphasis on the importance of distribution network construction, but the spatial resolution of the current study has not yet reached the level of detailed analysis at the distribution network level.

(5) National planning from a unified perspective. This study considers the whole country as a unity, without distinguishing specific regional differences, and therefore does not address the complexity of interregional transmission. This assumption is in line with the actual situation in China, considering that the planning, construction and post-completion scheduling of power sources in China are under the responsibility of a unified department.

(6) Uncertainty of new energy output. Considering the stochastic nature of the output of new energy sources such as wind and solar, this study adopts the confidence capacity of new energy sources as a proxy for their available installed capacity, which is estimated by the Equivalent Reliable Capacity Method.

4.2. Multi-objective solution based on NSGA-II coupled entropy weight TOPSIS

Scenario I with strong carbon constraints is set, and the forecast result of Scenario I in the electricity demand forecast in Chapter 2 is used as the electricity constraint, and the installed share of renewable energy reaches more than 60 % by 2030. Taking 2020 as the base year, the planning period runs from 2021 to 2030 (Sun Meng et al., 2023), and the planned power supply structure is the eight types of coal power and other power sources described in the previous section.

Addition of Base Carbon Constraint Scenario II, i.e., relaxation of the carbon emission limitation constraint of Eq. (7) and the installed capacity constraint of clean energy of Eq. (7) in Section 2.2, with the installed share of renewable energy reaching more than 50 per cent by 2030.

In order to explore the specific impacts of carbon constraints on the power structure, a control group scenario without carbon constraints is added for comparative analyses, which follows the power supply without carbon constraints in Chapter 3. In the power supply planning model presented in Chapter 3, Scenario 3 purposely excludes the carbon emission constraint in Eq. (7), thus ignoring the impact of environmental constraints on carbon emissions in the context of the ‘dual carbon’ target. Against the background of the current concern about air pollution, carbon dioxide emissions have become the focus of attention. In order to achieve the ‘dual carbon’ target within the specified time-frame, carbon emissions from the power sector need to be strictly controlled within the established targets. Carbon emissions from this sector are mainly due to the reliance on and consumption of fossil energy sources such as coal and oil by thermal power generation units. Specifically, this means that carbon emissions from coal-fired generating units in the electricity production process are not subject to the annual carbon emission targets.

The introduction of these two sets of control scenarios can reveal the difference between the development path of the power sector under no carbon constraints as well as under strong carbon constraints and when it achieves the carbon emission reduction target, and further emphasize the importance and urgency of implementing carbon constraints in promoting the power sector’s transition to a low-carbon regime and adjusting the power supply structure. The projection results of the three scenarios are shown in Table 7.

As can be seen from Table 7, in Scenarios 1 and 2, where carbon constraints are introduced, the share of renewable energy in total power generation in 2025 has reached 53.97 % and 51.4 %, respectively, which is significantly higher than Scenario 3, where no carbon constraints are introduced, where the share of renewable energy is 46.62 % in the same year, and only increases to 49.1 % in 2030. This comparison shows that considering carbon constraints in the power supply planning process can effectively increase the share of renewable energy in power generation while reducing the installed capacity of coal-based thermal power generation. On the contrary, when carbon emission limits are not considered, the lower initial investment and construction costs of thermal power generation, especially coal power, lead to a rapid increase in installed capacity. Therefore, by effectively managing and controlling carbon emissions when formulating power supply planning strategies, it not only promotes the development and application of renewable energy technologies, but also helps to achieve the optimization of the energy mix and the low-carbon transformation of the power system. Carbon constraints become a key factor in driving up the proportion of renewable energy and reducing dependence on fossil energy for power generation, which is of significant importance in reaching international carbon reduction targets and promoting environmental sustainability. In order to explore in detail China’s power supply planning for the period 2025–2030, this paper focuses on Scenario 1, which provides an in-depth discussion of measures to enhance the low-carbon transition.

The results of the selection of the Pareto frontier solution under scenario one, for power planning in 2030, are shown in Fig. 5.

By applying the entropy-weighted TOPSIS model for comprehensive

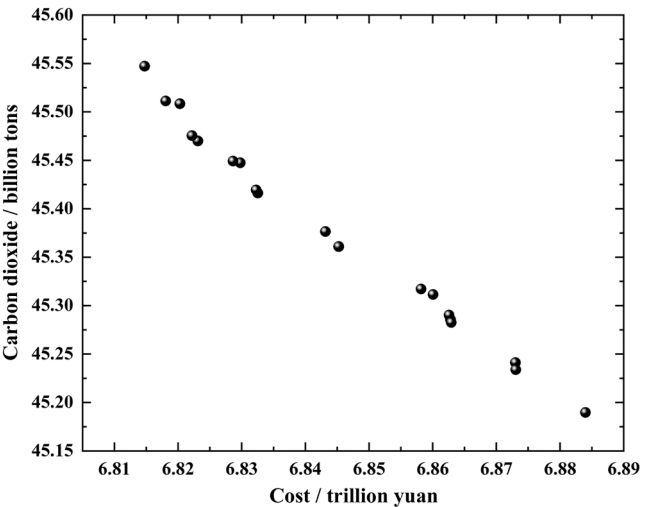


Fig. 5. Pareto optimal solution.

evaluation and screening, this paper identifies a compromise scenario under which the total cost required for the power structure is expected to be about RMB 6.86 trillion, with a corresponding CO₂ emission of 4.529 billion tons. This result coincides with the 4.6 billion tons predicted by the Chinese Academy of Engineering (CAE) for the peak carbon emission platform of the power system in 2030 (Shu et al., 2023), indicating that this study has high reliability and strong practical significance in power supply planning. This not only proves the validity of the selected compromise solution, but also emphasizes the importance of comprehensively considering the economic costs and environmental impacts in power supply planning, which provides a scientific basis and strategic support for realizing the low-carbon transition of the power system.

Figs. 6 and 7 present the evolution of the installed capacity and generation mix of power sources in China’s power system between 2021 and 2030. Up to 2030, while the installed capacity of coal power increases slightly, which is explained by the fact that coal power continues to play a key role in the total domestic power generation mix, it faces multiple challenges. Specifically, the cost of coal power is rising due to shrinking operating hours, rising coal costs, increased flexibility in the operation of coal units, and the phasing out of inefficient thermal generating units. As a result, in terms of the share of installed capacity and power generation, the share of coal power is showing a continuous downward trend. This changing trend reflects the fact that China’s power system is undergoing a shift from reliance on traditional coal power to a cleaner and more efficient energy mix. With the falling cost of renewable energy technologies and their increasing share in the power mix, as well as the pursuit of a low-carbon, environmentally friendly power supply, the position of coal power in China’s energy mix is expected to be further adjusted in the future. This process will not only be important for reducing carbon emissions and achieving the goal of carbon neutrality, but will also play a key role in promoting energy

Table 7
Power structure in 2025, 2030 under three scenarios (unit: 10 MW).

Power source	Scenario 1		Scenario 2		Scenario 3	
	2025	2030	2025	2030	2025	2030
coal power	116354.3	117543.3	116062.4	122839	131536.1	141390.8
gas power	16206.73	23470.13	14444.63	25771.7	26198.4	46294.01
hydropower	37974.97	41195.58	40823.31	47715.89	40232.85	46465.08
photovoltaic	42118.97	59737.19	28914.89	29779.61	29011.58	30601.89
Wind power	43033.32	61666.38	31842.62	35877.24	32814.69	44906.16
Nuclear power	11660.06	21997.97	19225.74	34804.18	19330.38	32779.3
Biomass power	12814.9	23088.9	11451.09	17696.44	10653.93	17417.11
pumped storage	7823.026	14038.5	5773.492	9341.275	5723.933	8913.493
share of new sources	53.97 %	61.13 %	51.40 %	54.10 %	46.62 %	49.10 %

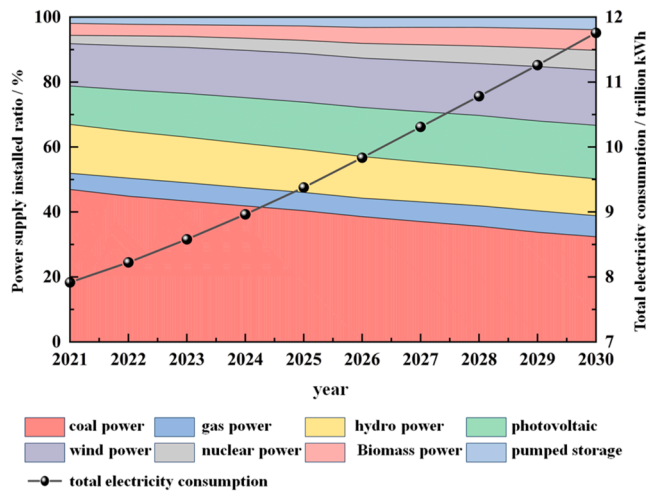


Fig. 6. Installed power supply structure 2021–2030.

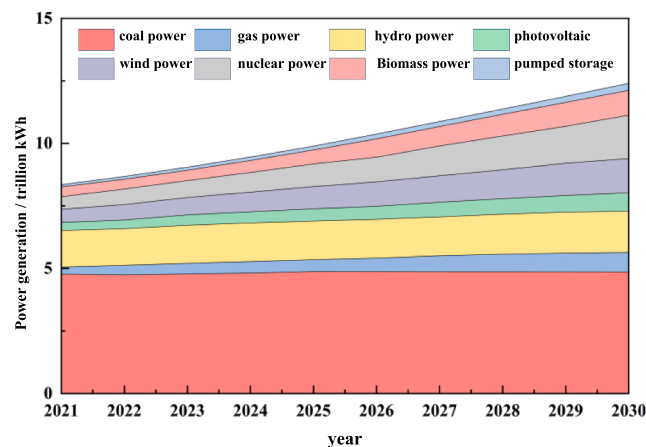


Fig. 7. Power generation share 2021–2030.

security, improving energy efficiency and fostering the development of a green and low-carbon economy.

Academician Y. Shu predicted in the literature (Shu et al., 2021) that the proportion of China's installed non-fossil energy capacity will reach 64 % by 2030. According to the power supply planning analysis in this study, the proportion of installed non-fossil energy is expected to rise to 61.13 % in 2030, which is similar to academician Shu's prediction. In addition, according to a State Council press release, China's total installed capacity of wind and solar energy is expected to reach 1.2 billion kW by 2030. As new energy technologies such as wind and solar continue to mature and develop, they will gradually become the dominant sources of energy in the power system. China has abundant potential in wind and solar resources, and is actively promoting the synergistic development of centralized and distributed power generation, optimizing the layout of the power structure, and accelerating the process of connecting new energy sources to the grid. The planning results in this paper show that the total installed capacity of wind and solar energy is expected to reach 1.214 billion kW, which is basically in line with the national target.

Despite the relatively small share of installed capacity in the current power sector, the potential of biomass for replacing traditional fossil fuels, especially petrol and diesel in transport and heating, cannot be ignored. In Scenario 1, the installed capacity of biomass power generation could reach 128 million kW in 2025 and 231 million kW in 2030, which not only provides a key negative emissions strategy for the power system, but also serves a variety of functions such as regulating grid

loads and providing district heating, making it an indispensable link in the optimization of power supply structures. The installed capacity of biomass power generation is expected to grow significantly as the technology continues to develop and become more economical.

Hydropower plays a crucial role as a central component of China's clean energy strategy. In pursuit of the dual carbon goals - i.e., carbon peaking and carbon neutrality - it is expected that conventional hydropower resources will continue to be exploited until they approach the upper limit of their maximum potential, i.e., 520 million kW (Liu et al., 2022a). According to the plan of this study, the installed capacity of hydropower in China will reach 412 million kW in 2030, and the full exploitation of hydropower resources is expected to be completed around 2060. Similarly, nuclear power, as a highly efficient and stable low-carbon energy source, is expected to gradually expand its development to the potential upper limit based on technological advances. However, given the public safety risks associated with nuclear energy development, China will take a cautious approach to nuclear power projects, prioritizing suitable locations such as coastal areas. This strategy aims to balance the economic and environmental benefits of nuclear power generation with potential safety risks to ensure the sustainability and security of energy development. In Scenario 1, the installed nuclear power capacity could reach 117 million kW in 2025 and 220 million kW by 2030.

Although the proportion of installed capacity of gas power and pumped storage is relatively low, it plays an important role in regulating the power system, which can effectively improve the system's ability to consume new energy and reduce the phenomenon of wind and light abandonment on the basis of ensuring the security and reliability of the system's power supply. Pumped storage is the main form of energy storage, according to the national target, that is, in 2030 to reach 120 million kW or so, in 2060 when the development of 420 million kW. By the results of this paper, the scale of commissioning in 2030 will reach 140 million kW, exceeding the national target.

5. Conclusions and Recommendations

This paper proposes a model for forecasting electricity demand under the carbon neutral target and a method for planning the electricity structure under this target, and verifies the validity of the model through vertical and horizontal comparisons.

- 1) A system dynamics model considering the carbon emission constraint is proposed, which effectively integrates the carbon emission constraint into the physical model of electricity demand forecasting, and the average absolute percentage errors are 1.73 % and 1.71 % through the model training and testing, and the model has a high forecasting accuracy. The prediction results show that the prediction result corresponding to Scenario 1 is 1.176×10^{13} kWh, which is 0.3 % different from the result of 1.18×10^{13} kWh given by the Chinese Academy of Engineering.
- 2) A dual-objective power supply planning model with total cost and carbon emission as the objective function is developed, and the NSGA-II coupled entropy weight TOPSIS model is used to effectively solve the dual-objective planning model with carbon constraints. On the basis of this projection, our goal of carbon neutrality can be achieved on schedule.

In addition, at present, China's power demand still shows an increasing trend, coal power is still the main power source, wind, light and other new energy power generation has a greater potential for improvement, based on the status quo, this paper puts forward the following suggestions:

- 1) Develop coal power in an orderly, clean and efficient way. Strictly control the total scale of coal power generation and the installation

of new units, give full play to the guiding and binding force of power planning, and implement the coordination, unity and orderly convergence of national planning and provincial planning. Comprehensive electricity demand and environmental carrying capacity, according to the need to promote the construction of coal-fired electricity, and gradually phase out long-life, energy inefficient thermal power units. Develop energy-saving renovation technologies for thermal power units, realize the gradual utilization of energy, reduce the coal consumption of power supply units, promote the popularization of new energy-saving technologies and techniques, and improve the advanced nature of the energy consumption indicators of newly-built projects.

- 2) Accelerate the substitution of clean energy. Increase the proportion of non-fossil energy power generation, actively develop wind power and photovoltaic power generation in an orderly manner, optimize the power supply structure, implement a renewable energy quota system, and include the proportion of clean energy power generation in the power generation evaluation index. Vigorously develop hydropower, promote the comprehensive utilization of hydropower in river basin terraces, and at the same time promote the construction of pumped storage power plants to enhance the flexibility of the power system operation, so as to provide a guarantee to cope with the uncertainty of renewable energy. Moderate development of gas-fired power generation, orderly construction of natural gas-fired peaking power plants, and development of gas-steam combined cycle power generation in accordance with heat load demand. Safe and efficient development of nuclear power, clear technology development paths to promote the process of commercialization and specialization, and the gradual formation of a nuclear fuel cycle system based in China and facing the world.

It is important to note that promoting the use of energy-saving retrofit technologies and renewable energy quota systems can encourage regions to vigorously promote the development of renewable energy sources such as wind power and photovoltaic power generation. For example, literature (Yuanyuan et al., 2022) indicates that by optimizing renewable energy quotas, it is possible to increase the installed capacity of wind power in non-hydropower regions from 31.9 % to 65.7 %. At the same time, hydrogen and thermal energy storage can reduce the cost of long-term and large-scale energy storage and are characterized by high efficiency and low or even zero carbon emissions (Zheng et al., 2024). Long-term power generation planning models integrating multiple energy sources should be developed in response to the fact that the structure of energy supply is undergoing significant restructuring as energy demand grows rapidly and investment budgets for energy systems may be more limited.

- 3) Control the “dual-carbon” goal and strengthen institutional mechanisms. Promote the construction of a national electricity market and improve the national carbon market system. Integrate and take into account the multi-level power market, improve and unify the trading guidelines and technical standards, establish standardized and unified power market price rules, develop capacity compensation mechanisms and ensure power system capacity adequacy. Establish a standardized carbon trading system and form a carbon trading market. Improve the carbon quota allocation system, perfect the integration mechanism between the power market and the carbon market, and promote the low-carbon development of the power system. Furthermore, in addition to centralized wind and photovoltaic power generation, regional power grids and renewable energy interconnections are developing more rapidly and are also elements to be considered (Jiaxi et al., 2024). Therefore, potential issues such as the security and stability of regional electricity systems and the allocation of electricity resources across regions should also be considered in the future.

CRediT authorship contribution statement

Li Changyun: Conceptualization. **Li Guangxiao:** Data curation. **Yang Yilu:** Formal analysis. **Liu Zongjie:** Funding acquisition. **He Zhaohui:** Resources, Methodology.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Changyun Li reports financial support was provided by Shandong University of Science and Technology. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- Abbasimehr, H., Shabani, M., Yousefi, M., 2020. An optimized model using LSTM network for demand forecasting. *Comput. Ind. Eng.* 143, 106435.
- Ahmad, T., et al., 2019. Nonlinear autoregressive and random forest approaches to forecasting electricity load for utility energy management systems. *Sustain. Cities Soc.* 45, 460–473.
- BASHIRI, B.N., et al., 2023. Incorporating air temperature into mid-term electricity load forecasting models using time-series regressions and neural networks. *Energy* 278, 127831.
- Bot, K., et al., 2020. Forecasting Electricity Demand in Households using MOGA-designed Artificial. *Neural Netw., IFAC Pap.* 53-2, 8228–8230.
- Brusaferri, et al., 2022. Probabilistic electric load forecasting through Bayesian Mixture Density Networks. *Appl. Energy* 309, 118341.
- Chen, W., Zhou, C., 2014. Evaluation of water resources carrying capacity of Wuhan city based on system dynamics simulation, *China Engineering Science*, 16 (03), 103–107, 112. (in Chinese).
- Deng, X., Lv, T., 2020. Power system planning with increasing variable renewable energy: A review of optimization models. *J. Clean. Prod.* 246, 118962.
- Fan, G., et al., 2021. Forecasting short-term electricity load using hybrid support vector regression with grey catastrophe and random forest modeling. *Uti. Policy* 73, 101294.
- Gao, Z., et al., 2022. A two-layer SSA-XGBoost-MLR continuous multi-day peak load forecasting method based on hybrid aggregated two-phase decomposition. *Energy Rep.* 8, 12426–12441.
- Jiaxi, L.I., Zhou, Zhuomin, Wen, Ming, et al., 2024. Multi-objective optimization method for medium and long-term power supply and demand balance considering the spatiotemporal correlation of source and load. *Energy Strategy Rev.* 54, 101463.
- Lei, H., Yao, X., Zhang, J., 2020. The competitiveness of provincial electric power supply in China: Based on a bottom-up perspective. *Int. J. Electr. Power Energy Syst.* 116, 105557.
- Li, J., et al., 2022. Long-term electricity consumption forecasting method based on system dynamics under the carbon-neutral target. *Energy* 244, 122572.
- Lindberg, K.B., et al., 2019. Long-term electricity load forecasting: Current and future trends. *Uti. Policy* 583, 102–119.
- Liu, S., et al., 2022a. Electric power supply structure transformation model of China for peaking carbon dioxide emissions and achieving carbon neutrality. *Energy Rep.* 8, 541–548.
- Liu, X., et al., 2022b. Improving carbon emission performance of thermal power plants in China: An environmental benchmark selection approach. *Comput. Ind. Eng.* 169, 108249.
- Luo, C., Zhang, K., 2024. Elasticity of substitution between clean energy and non-clean energy: Evidence from the Chinese electricity industry. *Cleaner Energy Systems*, 8, 100117.
- Ma, X., et al., 2024. Energy supply structure optimization of integrated energy system considering load uncertainty at the planning stage. *Energy* 305, 132187.
- Manesh, M.H.K., Davadgaran, S., Rabeti, S.A.M., 2024. New procedure for designing an innovative biomass-solar-wind polygeneration system for sustainable production of

- power, freshwater, and ammonia using 6E analyses, carbon footprint, water footprint, and multi-objective optimization. *Renew. Energy* 230, 120802.
- Scuzziato, M.R., Finardi, E.C., Frangioni, A., 2021. Solving stochastic hydrothermal unit commitment with a new primal recovery technique based on Lagrangian solutions. *Int. J. Electr. Power Energy Syst.* 127, 106661.
- Shu, Y., et al., 2021. Research on the carbon peaking and carbon neutral pathway of China's power. *China Eng. Sci.* 23 (06), 1–14.
- Shu, Y., et al., 2023. Low-carbon transformation path of China's energy and electricity under the "double carbon" target [J]. *Chin. J. Electr. Eng.* 43, 1663–1672.
- Sun Meng, M., et al., 2023. Analysis of key influencing factors and scenario projections of China's carbon emissions under the carbon neutrality target. *High. Volt. Technol.* 49 (9), 4011–4022 (in Chinese).
- Suneetha Emmela, et al., 2024. Global ionospheric total electron content short-term forecast based on Light Gradient Boosting Machine, Extreme Gradient Boosting, and Gradient Boost Regression. *Adv. Space Res.* 74, 3642–3653.
- Wan, B., et al., 2021. Green development growth momentum under carbon neutrality scenario. *J. Clean. Prod.* 316, 128327.
- Wang, B., et al., 2023. Multi-stage stochastic planning for a long-term low-carbon transition of island power system considering carbon price uncertainty and offshore wind power. *Energy* 282, 128349.
- Wang Zhengxin, Z., Qin, L.I., Lingling, P.E.I., 2018. A seasonal GM(1,1) model for forecasting the electricity consumption of the primary economic sectors. *Energy* 154, 522–534.
- Wei, F., et al., 2024. Carbon productivity growth in China's thermal power industry: empirical evidence using a bootstrapped Malmquist–Luenberger productivity index approach with fixed-sum undesirable outputs. *Energy* 304, 132034.
- Wu, Z., et al., 2024. Prediction model of electric power carbon emissions based on extended system dynamics. *Energies* 17, 472.
- WU, Y., et al., 2018. Short-term load forecasting based on improved grey correlation analysis with bat-optimized neural network. *Power Syst. Autom.* 42 (20), 67–72.
- Yao, X., et al., 2021. Low-carbon transformation of the regional electric power supply structure in China: A scenario analysis based on a bottom-up model with resource endowment constraints. *Resources. Conserv. Recycl.* 167, 105315.
- Yuan, K., et al., 2023. WExploration of low-cost green transition opportunities for China's power system under dual carbon goals. *J. Clean. Prod.* 414, 137590.
- Yuanyuan, H.E., et al., 2022. Regional renewable energy installation optimization strategies with renewable portfolio standards in China. *Sustainability* 14 (17), 10498.
- Zhang, M., et al., 2023. Accurate ultra-short-term load forecasting based on load characteristic decomposition and convolutional neural network with bidirectional long short-term memory model. *Sustain. Energy, Grids Netw.* 35, 101129.
- Zheng, L.I., et al., 2024. Potential of hydrogen and thermal storage in the long-term transition of the power sector: a case study of China. *Energy* 307, 132512.
- Zhou, B., et al., 2024. Optimization strategy of power purchase and sale for electricity retailers in a two-tier market. *Heliyon* 10, e26333.
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